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Role of bioavailability and protein binding of four anionic perfluoroalkyl substances in cell-based bioassays for quantitative *in vitro* to *in vivo* extrapolations

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ABSTRACT

Perfluoroalkyl substances (PFAS) are persistent and pose a risk to human health. High throughput screening (HTS) cell-based bioassays may inform risk assessment of PFAS provided that quantitative *in vitro* to *in vivo* extrapolation (QIVIVE) can be developed. The QIVIVE ratio is the ratio of nominal (C_{nom}) or freely dissolved concentration (C_{free}) in human blood to C_{nom} or C_{free} in the bioassays. Considering that the concentrations of PFAS in human plasma and *in vitro* bioassays may vary by orders of magnitude, we tested the hypothesis that anionic PFAS bind to proteins concentration-dependently and therefore the binding differs substantially between human plasma and bioassays, which has an impact on QIVIVE. Solid phase microextraction (SPME) with C18-coated fibers served to quantify the C_{free} of four anionic PFAS (perfluorobutanoate (PFBA), perfluorooctanoate (PFOA), perfluorohexane sulfonate (PFHxS) and perfluorooctane sulfonate (PFOS)) in the presence of proteins and lipid, medium components, cells and human plasma over five orders of magnitude in concentrations. The C18-SPME method was used to quantify the non-linear binding to proteins, human plasma and medium, and the partition constants to cells. These binding parameters were used to predict C_{free} of PFAS in cell bioassays and human plasma by a concentration-dependent mass balance model (MBM). The approach was illustrated with a reporter gene assay indicating activation of the peroxisome proliferator-activated receptor gamma (PPARγ-GeneBLAzer). Blood plasma levels were collected from literature for occupational exposure and the general population. The $QIVIVE_{nom}$ ratios were higher than the $QIVIVE_{free}$ ratios due to the strong affinity to proteins and large differences in protein contents between human blood and bioassays. For human health risk assessment, the $QIVIVE_{free}$ ratios of many *in vitro* assays need to be combined to cover all health relevant endpoints. If C_{free} cannot be measured, they can be estimated with the MBM and concentration-dependent distribution ratios.

1. Introduction

Per- and polyfluoroalkyl substances (PFAS) are used in commercial, industrial and military products for waterproofing, grease- and heat-resistance (Gluge et al., 2020). PFAS are persistent in the environment (Evich et al., 2022), which has led to diffuse worldwide contamination, e.g., in rainwater, that regularly exceed health-based advisory values (Cousins et al., 2022). Epidemiological investigations and toxicological studies have revealed associations between exposure to PFAS and human diseases, caused by endocrine disruption (Mokra 2021), reproductive (Calvert et al., 2021) and developmental toxicity (Rappazzo et al., 2017), neurotoxicity (Mariussen 2012) and immunotoxicity (DeWitt et al., 2009).

Traditional toxicology based on *in vivo* animal testing is expensive and time-consuming, making it impossible to generate safety information for thousands of commercial PFAS. High-throughput screening (HTS) cell-based bioassays proposed by the Tox21 and ToxCast program may inform risk assessment provided that quantitative *in vitro* to *in vivo* extrapolation (QIVIVE) can be developed for HTS (Bell et al., 2018; Sipes et al., 2017). *In vitro* concentration-response curves (CRC) and thereof derived effect concentrations can be converted by QIVIVE models into equivalent *in vivo* concentrations which are further compared to the human exposure levels or translated by reverse dosimetry to external *in vivo* oral exposure (Wetmore 2015).

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Currently, QIVIVE models use nominal concentrations (C_{nom}), but PFAS strongly bind to proteins and lipids in bioassays and blood, which will lower their bioavailability and hence the freely dissolved concentrations (C_{free}) in different ways in bioassays and blood. This problem might be more severe for the anionic PFAS due to their strong and specific binding to proteins.

Effect concentrations in HTS assays are typically based on C_{nom} because there is no routine method available to measure C_{free} in the 1536-well plates used by Tox21 and most reported C_{free} stem from 96-well plates (Huchthausen et al., 2020) or larger formats or were merely predicted (Proenca et al., 2021). Negligible-depletion solid phase microextraction (SPME) has been the choice for quantification of C_{free} of neutral organic chemicals for almost two decades (Heringa et al., 2004; Kramer et al., 2012), and the development of C18-coated fibers and SPME based on mass balances allowed the extension to ionizable chemicals in cell-based bioassay, as well as the experimental quantification of C_{free} of ionizable chemicals in complex bioassay media (Henneberger et al., 2019b; Huchthausen et al., 2020).

If C_{free} cannot be measured directly, it can be predicted with mass balance models (MBM) (Fischer et al., 2017; Proenca et al., 2021). Chemicals may specifically bind or non-specifically partition to medium components and cells, resulting in C_{free} lower than C_{nom} (Henneberger et al., 2019a; 2020). Distribution ratios of chemicals between protein and water ($D_{prot/w}$), lipid and water ($D_{lip/w}$), cell and water ($D_{cell/w}$), and their corresponding volumes ($V_{protein}$, V_{lipid} and V_{cell}) serve as input parameters in these MBM (Fischer et al., 2017). SPME has also been used to measure the binding and partition constants of ionizable chemicals to diverse biological materials and cells (Henneberger et al., 2019a). Anionic chemicals are particularly challenging because they show non-linear binding isotherms to proteins and protein-rich biomaterials (Henneberger et al., 2019a) and while the non-linearity of protein binding of diverse organic ions has been investigated, no such studies exist for anionic PFAS.

PFAS have been reported to cause numerous toxic effects *in vitro* and *in vivo* (Zeng et al., 2019). Specific interactions of PFAS with functional proteins dominate their molecular initiating events, including activation of receptor-mediated signaling pathways (Behr et al., 2020; Qin et al., 2020), interference of activities and functions of enzymes (Liu et al., 2020; Wang et al., 2014) and competitive binding with transport proteins (Ren et al., 2016). Nuclear receptors are universal in HTS testing because reporter gene technology makes the interaction of chemicals with functional proteins measurable. The peroxisome proliferator-activated receptor (PPAR) family comprises endogenous receptors of fatty acids and is involved in the homeostasis of lipid and glucose metabolism. Due to the structural similarity of PFAS and fatty acids, PPARs have been widely proven to be the specific targets for certain PFAS (Vanden Heuvel et al., 2006). A PPARγ reporter gene assay was selected as an example to show how to characterize the difference of C_{nom} and C_{free} of PFAS in HTS cell-based bioassays.

In vitro specific effects of PFAS can be linked to *in vivo* effects via QIVIVE. For example, Loizou et al. (2021) predicted *in vivo* benchmark doses (BMD) of perfluorooctanoic acid (PFOA) from *in vitro* concentration–response curves of the U.S. Environmental Protection Agency (EPA) HTS database using physiologically based pharmacokinetic (PBPK) model-reverse dosimetry. Typical PBPK models use a large group of parameters involving chemical absorption, distribution, metabolism and excretion processes (Loizou et al., 2021; Worley and Fisher, 2015). This greatly limited the application of QIVIVE for the assessments of large numbers of PFAS in screening type assessments. Sipes et al. (2017) developed an intuitive model, a ratio of maximum blood concentration *in vivo* ($C_{max,blood}$) to half-maximum effective concentration (AC50) directly from *in vitro* Tox21-HTS assay. The ratio > 1 indicated that the *in vitro* effect likely also occurred in *in vivo*. In its simplest form, a QIVIVE ratio can be calculated by dividing the plasma concentrations by the effective concentrations *in vitro* (Henneberger et al., 2021).

We selected perfluorobutanoic acid (PFBA), PFOA, perfluorohexane sulfonic (PFHxS), perfluorooctane sulfonic acid (PFOS) as representatives of acidic PFAS. They are fully deprotonated at physiological pH to the corresponding anions, perfluorobutanoate, perfluorooctanoate, perfluorohexane sulfonate and perfluorooctane sulfonate. They were selected because anionic surfactants are known for their non-linear binding to proteins. The four selected anionic PFAS are a special case of anionic surfactants and represent a wide range of hydrophobicity due to the differences in alkyl chain length and two different acid groups. Therefore, they serve for a proof of principle if existing knowledge of anionic organic chemicals can be translated to anionic PFAS.

Typically, the octanol–water partition coefficients (log K_{ow}) are used as measure of hydrophobicity, but they can only be predicted and not measured for surface-active anionic PFAS (Table 1). The octanol–water distribution ratio (log D_{ow}) is strongly dependent on pH and the concentration of counterions (Park et al., 2020) and the predicted log D_{ow} (pH 7.4) cover four orders of magnitude (Table 1). A more suitable predictor of hydrophobicity of surfactants is the distribution ratio between liposomes and water ($D_{lip/w}$). The experimental log $D_{lip/w}$ of PFBA is 1.48, which is considered to be hydrophilic, while other three PFAS with log $D_{lip/w} > 2$ are more hydrophobic (Table 2, data from Ebert et al. (2020), measured with equilibrium dialysis).

The protein binding of PFAS and partitioning to medium and cells, as well as C_{free} in the *in vitro* bioassay were studied by SPME using C18-coated fibers. The measured binding parameters were further used to predict C_{free} by MBM. Literature blood levels of PFAS of workers and residents living near fluorochemical plants (Bao et al., 2022; Gao et al., 2015) and the general population (Gao et al., 2019, https://www.cdc.gov/exposurereport/data_tables.html) were converted to C_{free} in blood with an adjusted blood-MBM using experimental plasma binding constants. An overview of the current approach can be found in Fig. 1. The cytotoxicity and specific effects of PFAS were characterized in a PPARγ reporter gene assay in agonistic and antagonistic model as a proof of principle. For a comprehensive risk assessment, a large set of responsive *in vitro* assays would need to be applied but this study is a demonstration of the importance and impact of nonlinear protein binding of anionic PFAS on QIVIVE.

2. Materials and methods

2.1. Materials

Four PFAS (purity > 98%) were used in this study. Perfluorobutanoic acid (PFBA, CAS. 375-22-4) was purchased from J&K. Perfluorooctanoic acid (PFOA, CAS. 335-67-1), perfluorohexane sulfonate potassium (PFHxS-K, CAS. 3871-99-6) and perfluorooctane sulfonate potassium (PFOS-K, CAS. 2795-39-3) were from Sigma Aldrich. All PFAS were dissolved in methanol as stock solution. Bovine serum albumin (BSA, CAS. 9048-46-8) was from Sigma-Aldrich, Rosiglitazone (CAS. 122320-73-4) and C18-coated fibers (CAT. 5728I-U) were purchased from Sigma-Aldrich. DC™ protein assay kit (CAT. 500-0114) was from Bio-red. GeneBLAzer® FRET Assay kit (LOT. 21972428) was from Invitrogen. GeneBLAzer PPARγ-UAS-bla 293H Cells (HEK293H) were purchased from Thermo Fisher. The GeneBLAzer assay medium contained 98% Opti-MEM (LOT.11058-021, Gibco), 2% charcoal-stripped FBS (cs-FBS, LOT. 2001639, Gibco) and 100 U/mL Penicillin-Streptomycin (LOT. 15140-122, Gibco). Solvents used in the assays were methanol (CAS. 67-56-1, Chemsolute), acetonitrile (CAS. 75-05-8, Chemsolute) and formic acid (CAS. 64-18-6, Honeywell).

2.2. Methods

2.2.1. Method development

Fibers with low C18-coating volume (5 mm coating length, 173 nL coating volume) were used for the hydrophobic PFOS as it showed very strong affinity for the fibers. PFOS samples were incubated at 37 °C using high-speed vortex shaker (DMS-2500, 1200 rpm around a 3.6 mm

Table 1
Predicted octanol-water partition coefficients (log K_{ow}) of the neutral and anionic species, octanol-water distribution ratios (log D_{ow}) at pH 7.4, and experimental data for binding to bovine serum albumin (BSA) of PFBA, PFOA, PFHxS and PFOS. The non-linear binding to BSA was described with a non-linear mechanistic combined binding/partitioning model, where K_d is the dissociation constant and C_{max} corresponds to the number of binding sites Eq. (14). log $K_{specific}$ Eq. (15) and log $D_{non\text{-}specific}$ Eq. (16) refer to the specific and non-specific binding to the BSA according to ν, the molar ratio of PFAS to BSA (mol_{PFAS}/mol_{BSA}, Eq. (13)). The measured bovine serum albumin (BSA) to water distribution ratios (log $D_{BSA/w}$) were also fitted as a function of the aqueous concentration C_w with a Freundlich-type model (Eq. (12)), where the slope corresponds to n_{Fr}-1 and the intercept to the Freundlich constant K_{Fr}.

Chemical	log K_{ow} (neutral species)	log K_{ow} (anionic species)	log D_{ow} (pH7.4)	Mechanistic combined binding/partitioning model					Freundlich-type model	
				K_d [μmol/L_w]	C_{max} [μmol/L_{BSA}]	C_{max} [mol/mol_{BSA}]	log $K_{specific}$ [L_w/L_{BSA}] $\nu \ll 1$	log $D_{non\text{-}specific}$ [L_w/L_{BSA}] $\nu > 1$	log $D_{BSA/w}$ [L_w/L_{BSA}] (C_w in μM)	R^2
PFBA	2.60[a]/2.95[b]	−3.76[b]	−1.22[c]	18.3	9.22×10^3	0.450	2.40	1.63	$\log D_{BSA/w} = -0.391 \times \log C_w + 2.80$	0.67
PFOA	5.68[a]/5.73[b]	−0.585[b]	1.58[c]	0.317	2.54×10^4	1.24	4.60	3.70	$\log D_{BSA/w} = -0.381 \times \log C_w + 4.32$	0.98
PFHxS	3.69[a]/5.29[b]	−1.77[b]	1.65[c]	7.44×10^{-2}	1.88×10^4	0.919	5.10	3.47	$\log D_{BSA/w} = -0.480 \times \log C_w + 4.32$	0.97
PFOS	5.77[a]	−0.153[b]	3.05[c]	6.24×10^{-3}	8.01×10^3	0.391	5.81	4.84	$\log D_{BSA/w} = -0.381 \times \log C_w + 4.91$	0.96

[a] log K_{ow} predicted average by OPERA from https://comptox.epa.gov.
[b] log K_{ow} predicted by COSMOtherm.
[c] log D_{ow} were from Park et al. (2020) and predicted by MarvinSketch from https://chemaxon.com/.

Table 2
Distribution ratios of PFBA, PFOA, PFHxS and PFOS between liposomes and water ($D_{lip/w}$) (Ebert 2020), cells and water ($D_{cell/w}$) (Eq. (11)) and plasma and water ($D_{plasma/w}$). log $D_{plasma/w}$ ($\nu < 1$) refer to the specific binding and were from experiments (Eq. (15)). log $D_{plasma/w}$ ($\nu < 1$) were also predicted by the mass balance model (MBM, Eq. (17)) with log $K_{specific}$ of BSA (Table 1) and log $D_{lip/w}$. Contributions of protein and lipid to the $D_{plasma/w}$ ($\nu < 1$).

	log $D_{lip/w}$ [L_w/L_{lip}]	log $D_{cell/w}$ [$L_w/L_{cell,prot+lip}$]	log $D_{plasma/w}$ [$L_w/L_{prot+lip}$] (experiment) at $\nu < 1$	log $D_{plasma/w}$ [$L_w/L_{prot+lip}$] (MBM) at $\nu < 1$	Contribution of protein binding	Contribution of lipid binding
PFBA	1.61[a]	2.56	2.12	2.36	98.3%	1.68%
PFOA	3.52	3.27	4.60	4.56	99.1%	0.867%
PFHxS	4.13	3.23	5.18	5.06	98.9%	1.12%
PFOS	4.89	4.19	5.96	5.77	98.7%	1.26%

[a] log $D_{lip/w}$ of PFBA was predicted with a linear relationship of log $D_{lip/w}$ and the number of fluorinated carbons of PFBA (Figure A1).

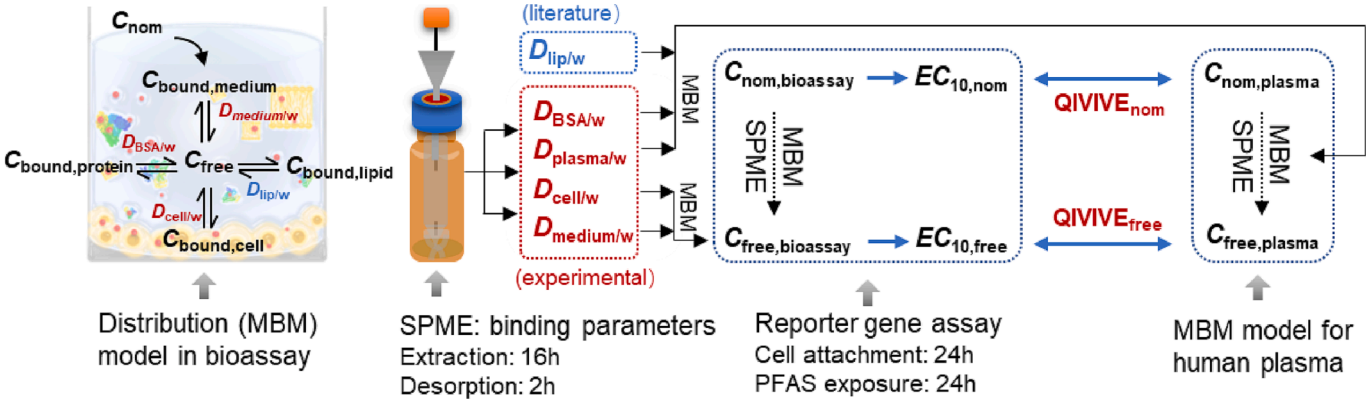


Fig. 1. An overview of the models and experiments needed to derive quantitative *in vitro* to *in vivo* extrapolation ratios (QIVIVE ratio) based on free and nominal (nom) concentrations (C_{free} and C_{nom}). Abbreviations: BSA = bovine serum albumin; $D_{i/w}$ = distribution ratio between i and water; EC = effect concentration; lip = liposome; MBM = mass balance model; SPME = Solid phase microextraction.

orbit) to accelerate adsorption. Pure solvent (100% MeOH) was used to desorb PFOS from fibers. For moderately hydrophobic PFOA, PFHxS and hydrophilic PFBA, fibers with a larger capacity (15 mm coating length, 520 nL coating volume) and a less vigorous shaking on an orbital shaker (MaxQ 6000, 250 rpm around a 19 mm orbit) was used for adsorption, and a mixed solvent (50% MeOH and 50% MilliQ water) was used for desorption. A summary of SPME method is given in Table A1.

There were three steps in the SPME process: adsorption, desorption and sample measurement. According to Henneberger et al. (2019a), C18-coated fibers were conditioned in methanol (2 h) and water (0.5 h) before being inserted to the samples.

2.2.2. Fiber uptake kinetics and sorption isotherm

Methanolic PFAS stock solutions were diluted with phosphate buffered saline (PBS). The concentrations of each PFAS are listed in Table A2. The final percentages of MeOH in samples were < 1%. The pH value of samples was adjusted to pH 7.4 using sodium hydroxide (NaOH) and hydrogen chloride (HCl) before SPME.

For adsorption kinetics, 18 aliquots of 1 mL for each PFAS solution were filled into 1.5 mL HPLC vials (ART. 7,654,554 or 7663230, Labsolute). The initial concentrations of each PFAS are listed in Table A2. The fibers were inserted into the vials individually using blunt cannulas. The samples were incubated at 37 °C using high-speed vortex shaker (1200 rpm) or orbital shaker (250 rpm) for 0.25, 0.5, 1, 2, 4, 6, 24, 48 h.

For sorption isotherm measurements, PFAS stock solutions were diluted with PBS serially with a factor of 2. The concentrations of each PFAS varied by 2–3 orders of magnitude. The concentration ranges are listed in Table A2. Duplicates of 1 mL for each concentration were filled into 1.5 mL HPLC vials. All the samples were shaken with the fibers for 16 h to reach equilibrium for adsorption.

The desorption and measurement were described in 2.2.4. Concentrations of PFAS in PBS and solvent extracts were used to calculate the recovery of the SPME method and distribution ratios of PFAS between fiber and water.

2.2.3. Partitioning to biological materials

The SPME measurement for PFAS partitioning to biological materials (BSA, plasma, medium, cells) were similar as above but different in sample preparations. For determination of protein-water distribution ratios, BSA was dissolved in PBS at a concentration of 5 g/L to aim at a fraction of PFAS bound to protein > 20%. The highest concentrations for PFBA, PFOA, PFHxS and PFOS were 2000, 500, 500 and 166 μmol/L and these concentrations were serially diluted by BSA buffer with a dilution factor of 2 to obtain 10 concentration points (Table A2). The concentration ranges of PFAS were designed to include both concentrations below and above saturation of the primary binding site as ν values (mol_{PFAS} /mol_{BSA}) ranged from 0.01 to 2. All the samples were shaken for 15 min (37 °C, 250 rpm) before aliquoting duplicates of 1 mL PFAS solution into 1.5 mL HPLC vials. The fibers were inserted into vials individually and the samples were shaken for 16 h to reach equilibrium between SPME fibers and all other components.

For determination of plasma-water distribution ratios, human plasma was diluted by PBS buffer to 20% or 4% plasma contents. PFAS solutions were serially diluted in PBS with a dilution factor of 2 to obtain 5 concentration points. 100 μL plasma buffer were transferred to HPLC vials with insert and 100 μL PFAS solution were added. The final plasma contents were then 10% or 2% and highest concentrations of PFBA, PFOA, PFHxS and PFOS were 500, 125, 125 and 50 μmol/L (Table A2). Samples were vortexed for 5 s to mix and then shaken for 15 min (37 °C, 250 rpm) before SPME.

For determination of medium-water distribution ratios, the highest concentrations for PFBA, PFOA, PFHxS and PFOS were 4000, 500, 500 and 200 μmol/L (Table A2) and GeneBLAzer assay medium was used to serially dilute PFAS in a 96-well plate with a dilution factor of 2 to obtain 10 concentration points. The bioassay medium contained 2% cs-FBS, which had proteins and lipids as sorptive phases, and the charcoal stripping (cs) was used to remove hormones and other compounds that would cause a background effect in the bioassay. Nominal PFAS concentrations ranged over 2–3 orders of magnitude while the contents of protein and lipid were constant in the experiments. 180 μL of PFAS solution were transferred from each well to HPLC vials with insert (CAT. 7648146 or 7651116, Labsolute). All the samples were shaken for 15 min (37 °C, 250 rpm) before SPME fibers were inserted.

For determination of cell-water distribution ratios, HEK293H cells were harvested by trypsinization and the number of cells was counted by CASY Counter (Roche Innovatis, Reutlingen, Germany) before centrifugation for a cell pellet. The cell pellet was resuspended with PBS at the density of 2×10^7 cells/mL. 1 mL cell suspension was homogenized by ultrasonic treatment (Sonoplus 2070, Germany) in an ice-water bath for 3×30 s to avoid protein denaturation due to localized overheating. 100 μL of cell homogenate were transferred to HPLC vials with insert and another 100 μL PFAS solution in PBS were added. Samples were vortexed for 5 s to mix and then shaken for 15 min (37 °C, 250 rpm) prior to inserting the SPME fibers.

2.2.4. Desorption and quantification of PFAS

For desorption, fibers were transferred to new vials with 100% MeOH (PFOS) or a mixture of 50% MeOH and 50% MilliQ water (PFOA, PFHxS, PFBA). The volume of samples and desorption solvents for different assays are listed in Table A3. Then, these vials were shaken with vortex or orbital shakers for another 2 h.

PFAS in PBS and all extracts were analyzed by a 1260 Infinity liquid chromatograph (LC) equipped with a Kinetex 1.7 μm, C18, 100 Å, LC column (50 × 2.1 mm) and coupled to 6420 Triple Quad mass spectrometer (MS) from Agilent. Detailed information of LC/MS analysis can be found in Table A4. All samples were diluted to concentrations in the calibration range of PFAS (1–10000 ng/L).

2.2.5. Lipid and protein quantification

Lipid and protein content of medium, cell homogenates, human plasma was determined according to Fischer et al. (2017). Protein and lipid content were quantified with protein assay kit and sulpho-phospho-vanillin reaction, respectively. Samples were diluted with MilliQ water to assure that the protein and lipid contents were within calibration ranges.

2.2.6. PPARγ reporter gene assay

The commercially available GeneBLAzer PPARγ-UAS-bla 293H cells (ThermoFisher) were cultured as described previously (Neale et al., 2017). There were four steps in GeneBLAzer Assay: cell attachment, PFAS exposure, cytotoxicity measurement and reporter protein detection.

2.2.6.1. Cell attachment. HEK293H Cells were harvested by trypsinization and the number of cells was counted by CASY Counter. Cell pellet was resuspended with assay medium at the density of 2.1×10^5 cells/mL. 100 μL of the cell suspension were seeded in each well of the 96-well plates (Corning® BioCoat™ Poly-D-Lysine). After the seeding, the plates were left to rest for 10–30 min at room temperature and incubated at 37 °C and 5% CO_2 and 100% humidity for 24 h to let the cells attach to the plate before dosing with PFAS.

2.2.6.2. PFAS exposure. PFAS stocks were prepared in MeOH. MeOH was evaporated with a gentle stream of nitrogen, leaving only PFAS precipitaties at the bottom of dosing vials. Assay medium contained 2% cs-FBS, which has a low background signal caused by hormone (data not shown), and was added to dissolve the PFAS before being serially diluted with medium in a 96-well plate. 120 μL solution from each well were transferred to previous 96-well plate with adhered cells using a 12-channel pipette. For antagonistic assay, medium containing rosiglitazone (agonist of PPARγ) was used to prepare PFAS solution (Table A2). The dilution process of PFAS was the same as above. The final concentration of rosiglitazone in the cell plate was 7 nmol/L, which was expected to trigger approximately 80% of the maximum effects. An 80% effect level was chosen following the recommendations of Neale et al. (2015) and because some chemicals may have potential weak activation effects at low concentrations. Both agonist and antagonistic effect could be recorded. After PFAS dosing, cell plates were incubated for another 24 h.

2.2.6.3. Cytotoxicity measurement. Cells were imaged non-invasively with the IncuCyte S3 (Essen BioScience) before dosing and the confluency of the cells was calculated by software (IncuCyte S3 v2019A) with details provided in Escher et al. (2019). The confluency of the cells

was measured again by IncuCyte S3 24 h after dosing. The cell viability was determined by comparing the confluency of the exposed and unexposed cells 24 h after dosing.

2.2.6.4. Reporter protein detection. Before the detection of reporter protein, 200 μL of supernatant from each well were transferred to a new 96-well plate and then used for SPME measurement (2.2.3). The activation signal of PPARγ coupled to a reporter gene encoding for β-lactamase was measured by fluorescence after addition of the GeneBLAzer® FRET reagent and additional 2 h incubation. The detailed protocol can be found in Neale et al. (2017).

2.3. Data evaluation

2.3.1. Fiber uptake kinetics

PFAS on the C18-coated fiber were desorbed by solvents and the extract was measured directly by LC/MS. The concentration of PFAS in the fiber (C_f) was calculated from the concentration measured in the extract ($C_{extract}$) and the volume of extract ($V_{extract}$) divided by the volume of fiber coating (V_f) (Eq. (1)).

$$C_f = \frac{C_{extract} \times V_{extract}}{V_f} \quad (1)$$

Adsorption kinetics determine the time required to reach equilibrium. In a single adsorbate system, the rate of adsorption process is expressed based on the amount of chemicals. The amount of PFAS in water as a function of time $n_w(t)$ can be defined by Eq. (2):

$$n_w(t) = n_w(eq) \times \left(1 - e^{-k_1 \times t}\right) + n_w(t_0) \times e^{-k_1 \times t} \quad (2)$$

$n_w(eq)$ and $n_w(t_0)$ are the amounts of PFAS in water at equilibrium and at the start of the experiment. k_1 is the rate constant for kinetic model. Since $n_f(t_0) = 0$, the amount of PFAS in fiber $n_f(t)$ can be fitted by Eq. (3):

$$n_f(t) = n_f(eq) \times \left(1 - e^{-k_2 \times t}\right) \quad (3)$$

The time to reach 95% of equilibrium of fiber absorption ($t_{0.95}$) was derived by Eq. (4).

$$t_{0.95} = \frac{\ln 0.05}{-k_2} \quad (4)$$

The recovery of PFAS during the SPME extraction was calculated by Eq. (5).

$$\text{recovery ratio} = \frac{n_f + n_w}{n_{tot}} \quad (5)$$

2.3.2. Sorption isotherms to the SPME fiber

Sorption isotherms can be used to describe the binding of chemicals to surfaces. The binding of anionic chemicals to C18 fibers is an adsorption process with limited binding sites. In this case the sorption isotherms of C_f against the aqueous concentration C_w are non-linear and experimental sorption data can be fitted with an empirical model, known as Freundlich isotherm (Eq. (6)).

$$C_f = K_{Fr} \times (C_w)^{n_{Fr}} \quad (6)$$

K_{FR} and n_{FR} are Freundlich constant and exponent, which can be deduced from experimental data by the logarithmic form of Eq. (6), which is (Eq. (7)):

$$\log C_f = n_{Fr} \times \log C_w + \log K_{Fr} \quad (7)$$

The apparent distribution ratio of PFAS between fiber and water $D_{f/w}$ Eq. (8) at equilibrium is dependent on the concentrations and can be calculated with K_{FR} and n_{FR} Eq. (9).

$$D_{f/w} = \frac{C_f}{C_w} \quad (8)$$

$$\log D_{f/w} = \left(1 - \frac{1}{n_{Fr}}\right) \times \log C_f + \frac{\log K_{Fr}}{n_{Fr}} \quad (9)$$

2.3.3. Binding of PFAS to biological materials

SPME measurements were used to quantify the aqueous concentration in presence of biomaterials (i = proteins, lipid, human plasma, medium or HEK293H cells) and the mass balance $n_{tot} = n_f + n_w + n_{bound}$ was used to derive the distribution ratios of PFAS between biomaterials i and water. Concentration of PFAS bound to sorptive phases i ($C_{bound,i}$) can be calculated by Eq. (10):

$$C_{bound,i}\left[\frac{\mu mol}{L}\right] = \frac{n_{bound,i}}{V_i} = \frac{n_{tot} - n_f - n_w}{V_i} \quad (10)$$

The distribution ratios of PFAS between phase i and water ($D_{i/w}$) were calculated from experimental data by Eq. (11) for each concentration and experiment.

$$D_{i/w}\left[\frac{L_w}{L_i}\right] = \frac{C_{bound,i}}{C_w} = \frac{\left(\frac{n_{tot}}{n_f} - 1\right) \times D_{f/w} \times V_f - V_w}{V_i} \quad (11)$$

The experimental concentration-dependence of $D_{i/w}$ was fitted with a Freundlich-type model using Eq. (12). The distribution to biomaterials, unlike binding to the SPME fiber, is not a pure surface sorption process but a mix of specific binding to proteins and non-specific partitioning to proteins and lipids. Therefore, Eq. (12) is only a phenomenological fit of the experimental data, and in case of n_{Fr} deviated from 1, a mechanistic model was invoked to describe the non-linear binding. However, the empirical model of the non-linear binding will suffice for developing a mass balance model for the bioassays and to derive the dissolved concentration C_w.

$$\log D_{i/w} = (n_{Fr} - 1) \times \log C_w + \log K_{Fr} \quad (12)$$

2.3.4. Mechanistic model for protein/plasma binding

Protein binding is known to be concentration-dependent and therefore $D_{BSA/w}$ was also modeled with a mechanistic combined binding/partitioning model. PFAS may bind specifically to one or more specific binding sites on proteins at low concentration and partition non-specifically to the protein and lipid at high concentrations. At a ratio ν of bound PFAS molecules to protein molecules (eq. (13), [$mol_{PFAS}/mol_{protein}$]) of $\nu < 1$, a saturation binding model (Eq. (14)) can be used to describe the binding curves of the bound concentration (C_{bound}) as a function of C_w with the dissociation constant K_d and C_{max} as maximum bound concentration. C_{max} reflects the number of binding sites on the protein in case of BSA after the concentrations in [$mol_{PFAS}/L_{protein}$] were converted to [mol_{PFAS}/mol_{BSA}] using the molecular weight of BSA of 66463 Da and a density of 1.36 kg/L (Eq. (13)).

$$\nu = \frac{n_{bound}}{n_{protein}} \quad (13)$$

$$C_{bound,specific} = \frac{C_{max} \cdot C_w}{K_d + C_w} \quad (14)$$

The low-concentration linear range of the binding curves extends approximately to $C_{max}/2$ and can be described by $K_{specific}$ (Eq. (15)).

$$K_{specific} = \frac{C_{max}}{2 \times K_d} \quad (15)$$

After obtaining the K_d and C_{max} from the low-concentration portion of the binding curve ($\nu < 1$) the non-specific distribution constant $D_{non-specific}$ was fitted to the full sorption model by keeping the previously determined K_d and C_{max} constants (Eq. (16)).

$$C_{bound,total} = \frac{C_{max} \cdot C_w}{K_d + C_w} + D_{non-specific} \times C_w \quad (16)$$

2.3.5. Mass balance model for distribution to biomaterials

The biomaterial-water distribution ratio $D_{i/w}$ can also be predicted by a mass balance model (MBM, Eq. (17)) from the concentration-dependent protein binding constants and the concentration-independent distribution to the lipid phases, assuming that protein and lipid are the major sorptive phases in the tested biomaterials (medium, cells, plasma).

$$D_{i/w} = D_{BSA/w} \times \frac{V_{prot}}{V_{prot+lip}} + D_{lip/w} \times \frac{V_{lip}}{V_{prot+lip}} \quad (17)$$

The volume of protein (V_{prot}) and lipid (V_{lip}) in plasma, medium, HEK293H cells were measured for the materials used in this study as described by Fischer et al. (2017).

$D_{BSA/w}$ measured here served as proxy for protein distribution and the liposome (phospholipid vesicles)-water distribution ratio $D_{lip/w}$ from Ebert et al. (2020) as proxy of lipid distribution. The $D_{lip/w}$ of PFBA was predicted from a linear relationship between log $D_{lip/w}$ of perfluorocarboxylic acids and the number of fluorinated carbon (C-F) (Figure A1). The R-squared of this linear regression was 0.95, the slope was 0.476 and intercept was 0.172.

2.3.6. Freely dissolved concentration in the bioassays

The aqueous concentration in the bioassay medium was quantified at the end of the bioassay experiment. Since the SPME extraction leads to a substantial depletion of some of PFAS, one needs to back-calculate to the original aqueous concentration in the bioassay using Eq. (18) (Henneberger et al., 2019b). In *in vitro* toxicology, this aqueous concentration is often termed freely dissolved concentration ($C_{free,bioassay}$) and we use this terminology for consistency. The $D_{f/w}$ used for this conversion was calculated from C_f using the equations given in Table A6.

$$C_{free,bioassay} = \frac{n_{tot}}{D_{f/w} \times \left(\frac{n_{tot}}{C_f} - V_f\right)} \quad (18)$$

$C_{free,bioassay}$ can also be predicted by a mass balance model (Eq. (19)) with the $D_{i/w}$ of different biomaterials i (medium, cells) and their corresponding volume V_i.

$$n_{tot} = n_{free} + n_{bound,i} = C_{free,bioassay} \times \left(V_w + \sum_{i=1}^{n} (D_{i/w} \times V_i)\right) \quad (19)$$

The $C_{free,bioassay}$ can be calculated by rearranging Eq. (19) and introducing the measured $D_{medium/w}$ of GeneBLAzer assay medium and $D_{cell/w}$ of the HEK293H cell line (Eq. (20)).

$$C_{free,bioassay} = \frac{n_{tot}}{(D_{cell/w} \times V_{cell,\ prot+lip} + V_{cell,\ w}) + (D_{medium/w} \times V_{medium,prot+lip} + V_{medium,w})} \quad (20)$$

2.3.7. PPARγ bioassay

Cytotoxicity was expressed as a percentage of reduced cell numbers (confluency) compared to the unexposed cells (eq. (21)). The concentration-cytotoxicity curve, which is typically linear up to 30% reduction of confluency (Escher et al., 2018) was fitted with Eq. (21).

$$\%\text{Cytotoxicity} = 100\% - \frac{\text{confluency(sample)}}{\text{confluency(unexposedcells)}} = \text{slope}_{cytotoxicity} \times \text{concentration} \quad (21)$$

The inhibitory concentration that caused 10% of cytotoxicity (IC_{10}) was calculated from the linear portion of the concentration-cytotoxicity curve below 30% cytotoxicity with Eq. (22) (Escher et al., 2018).

$$IC_{10} = \frac{10\%}{\text{slope}_{cytotoxicity}} \quad (22)$$

The specific effect was the reporter gene activation or suppression of PPARγ. For agonistic effects, effective concentration that triggered 10% of the maximum PPARγ activation induced by the positive control rosiglitazone (EC_{10}, Eq. (23)) was calculated from the linear portion of concentration-effect curve (eq. (23)), at non-cytotoxic concentrations, i.e., at concentrations below IC_{10} and below 30% of maximum effect, with Eq. (24).

$$EC_{10} = \frac{10\%}{\text{slope}_{effect}} \quad (23)$$

$$\%\ \text{of}\quad \text{maximum}\quad PPAR\gamma\quad \text{activation} = \text{slope}_{effect} \times \text{concentration} \quad (24)$$

For antagonistic effects, 7 nM rosiglitazone (agonist of PPARγ) was added in all wells to trigger approximately 80% of the maximum effects and a dilution series of the PFAS was run against that background. The suppression ratio (SPR) of PFAS can be calculated by the reduction of the activation signal of the constant concentration of rosiglitazone (Eq. (25)).

$$SPR = 100\%\text{-}\frac{\text{Activation}_{sample}}{\text{Activation}_{rosiglitazone}} \quad (25)$$

The effective concentration causing a SPR of 20% (EC_{SPR20}, Eq. (26)) was derived from the linear portion of the concentration-suppression curve (similar to Eq. (25)).

$$EC_{SPR20} = \frac{20\%}{\text{slope}} \quad (26)$$

3. Results and discussions

3.1. Fiber uptake kinetics and sorption isotherm of PFAS

To establish a SPME method specifically for PFAS, we performed kinetic uptake and sorption isotherm experiments (Figure A2) to study adsorption characteristics of PFAS to C18-coated fibers. All four PFAS reached equilibrium status within 16 h (Eq. (4), Fig. A2a-d and Table A1) and their average recovery ratios (Eq. (5)) were 95%-120%. Hydrophobic PFHxS, PFOA and PFOS showed strong affinities to fibers in a concentration-dependent way. The sorption isotherm was linear for PFBA but non-linear for the other PFAS with Freundlich exponents n_{Fr} ranging from 0.641 to 0.764 (Eq. (7), Fig. A2e-h, Table A6). The $D_{f/w}$ of PFHxS, PFOA and PFOS hence showed a strong concentration dependence (Eq. (9), Fig. A2j-l and Table A6).

3.2. Binding to proteins

The concentrations of PFAS bound to BSA ($C_{bound,BSA}$) (Eq. (10)) increased non-linearly with C_w (Fig. 2a-d). In the experiment, the volume of BSA was kept constant, while the concentration of PFAS ranged over four to five orders of magnitude. The binding curves of PFOA, PFHxS and PFOS showed three phases. The PFAS appeared to bind to the high-affinity sites of BSA at low concentrations. When these sites gradually saturated, a transition stage in the middle part of curve appeared. At high concentrations, non-specific partitioning was observed. For hydrophilic PFBA, the transition phase was affected by non-specific binding. The binding curves

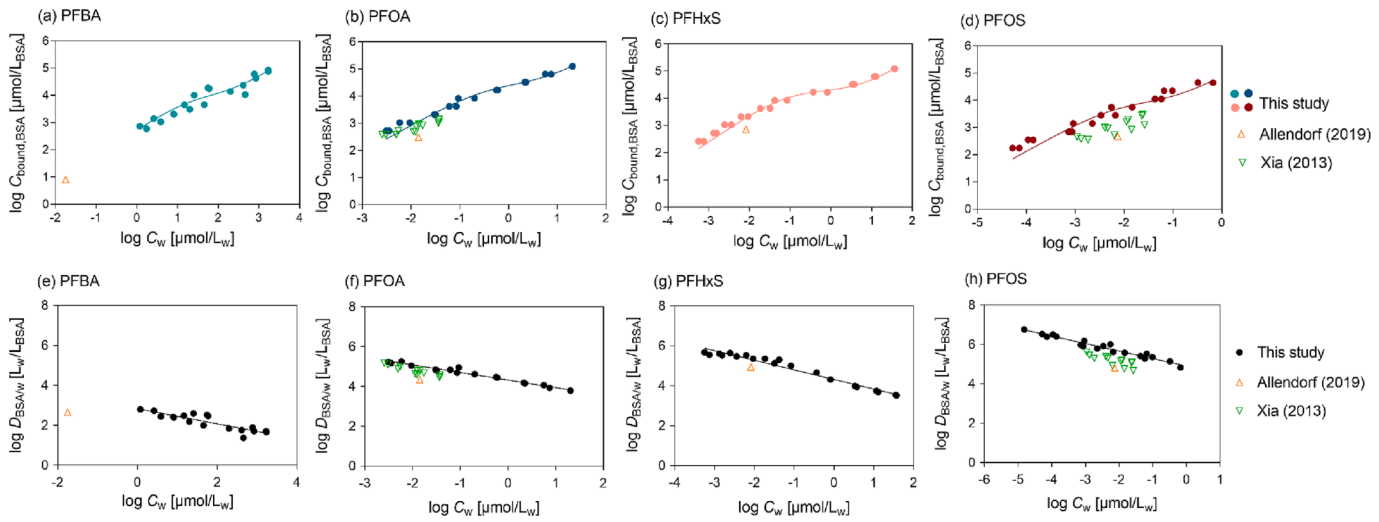


Fig. 2. Protein (BSA) binding curves. The concentrations of (a) PFBA, (b) PFOA, (c) PFHxS and (d) PFOS bound to bovine serum albumin ($C_{bound,BSA}$, circles) were fitted with a mechanistic combined binding/partitioning model (Eq. (16)). Distribution ratios between BSA and water ($D_{BSA/w}$) for (e) PFBA, (f) PFOA, (g) PFHxS and (h) PFOS. log $D_{BSA/w}$ (black circle) was fitted phenomenologically by a Freundlich-type model (Eq. (12)). The experimental data were compared with previous dialysis experiments from Allendorf et al (2019) (orange upward triangles) and Xia et al (2013) (green downward triangles) for comparison. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

were fitted with Eq. (16). The specific saturation binding $K_{specific}$ (Eq. (15)) and the non-specific distribution ratio $D_{non\text{-}specific}$ (Eq. (16), Table 1) were not as distinctly differentiated as for other anionic organics such as naproxen (Henneberger et al., 2019a). The C_{max} of PFBA, PFOA, PFHxS and PFOS were 0.450, 1.24, 0.919 and 0.391 mol_{PFAS}/mol_{BSA}, respectively (Table 1), indicating that there is most likely only one high-affinity binding site for PFAS. PFOS showed the highest potency of specific protein binding with a log $K_{specific}$ of 5.81. The log $D_{non\text{-}specific}$ ($\nu > 1$) (Table 1) were lower than log $K_{specific}$ but were remarkably similar to the log $D_{lip/w}$ for PFBA, PFOA and PFOS (Table 2).

Since PFAS binding to BSA was non-linear, $D_{BSA/w}$ (Eq. (11)) of PFAS were concentration-dependent (Fig. 2e-h). The decrease of experimental $D_{BSA/w}$ with increasing C_w was phenomenologically fitted with Eq. (12) and the resulting Freundlich parameters K_{Fr} and n_{Fr} are listed in Table 1. Note that this very simplified linearization of the three-phasic binding curve does not reflect the complex binding mechanism, but the resulting linear regressions may serve as input function for the MBM.

There are two studies using dialysis experiments to investigate the protein binding of PFAS. Allendorf et al. (2019) measured BSA-water partition coefficients for seven PFAS at one constant concentration. Their experimental data of PFBA, PFOA, PFHxS and PFOS are compared in Fig. 2 with the present binding curves measured with SPME. All are situated at different regions of the binding curves. PFBA (Fig. 2a, orange upward triangle) was measured at very low linear specific binding at concentrations that were two orders of magnitude lower than the concentration range measured here, PFOA (Fig. 2b, orange upward triangle) and PFHxS (Fig. 2c, orange upward triangle) were situated at the linear specific binding concentration range, and PFOS (Fig. 2d, orange upward triangle) at the transition zone, where specific binding was going towards saturation and non-specific binding contributed to the overall binding.

Xia et al. (2013) studied the protein binding of six PFAS with different protein concentrations of BSA and found non-linear binding curves of PFOA and PFOS. Their binding curves for PFOA (Fig. 2b, green downward triangles) were in the range of specific binding and their experimental data from dialysis experiments agreed remarkably well with our data from SPME experiments. PFOS binding was measured at the same concentration range as PFOA but fell into the transition zone between saturation of specific binding and non-specific binding (Fig. 2d, green downward triangles). The present SPME experiment indicated stronger binding in the same concentration range as the dialysis experiment data from Allendorf et al. (2019) and Xia et al. (2013).

Bischel et al. (2010) found up to eight binding sites in PFAS-BSA complexes at $\nu > 4$ (mol_{PFAS}:mol_{BSA}) using Nano-electrospray ionization mass spectrometry. Human serum albumin (HSA) and BSA are 76% conserved in sequences (Theodore 1985) and thus some PFAS showed similar affinity to both BSA and HSA (Bischel et al., 2010). Chen and Guo (2009) further investigated the affinities of PFAS on three binding sites of HSA using site-specific fluorescence. PFBA, PFOA, PFOS had different preferences among these binding sites with orders of magnitude difference in binding constants. If the strongest binding site contributed the most of the PFAS binding to protein, the binding model can be simplified as one high-affinity binding site model as we did here.

Several models have been used in previous studies to analyze how PFAS-protein binding is affected by hydrophobicity, steric limitation and number of fluorinated carbon ($n_{C\text{-}F}$) (Alesio et al. 2022; Almeida et al., 2021; Qin et al., 2020; Ren et al., 2016). Their results suggested that $n_{C\text{-}F}$ plays an important role in PFAS-protein binding because it has positive correlations with chemical hydrophobicity and molecular size (Alesio et al., 2022). In our study, the value of log $D_{non\text{-}specific}$ showed a good relationship with $n_{C\text{-}F}$, PFOS ($n_{C\text{-}F}$:8) > PFOA ($n_{C\text{-}F}$:7) > PFHxS ($n_{C\text{-}F}$:6) > PFBA ($n_{C\text{-}F}$:3), suggesting $n_{C\text{-}F}$ may dominate the non-specific binding of PFAS to protein (Figure A3). Different functional groups (e.g., sulfonic, carboxylic or hydroxy group) were also reported to impact the interactions of PFAS with protein, such as van der Waals interactions, electrostatic interactions and hydrogen bonding (Almeida et al., 2021; Qin et al., 2020; Ren et al., 2016). The higher values of log $K_{specific}$ (Table 1) of PFHxS and PFOS suggest that sulfonic acids may have stronger binding affinity to the ligand binding pockets, which is consistent with previous studies using nuclear protein (e.g., PPARγ), membrane protein (e.g., GPR40) and transport protein (e.g. transthyretin) (Almeida et al., 2021; Qin et al., 2020; Ren et al., 2016).

Overall, the comparison with literature data demonstrates that it is important to measure protein binding at a wide concentration range to assure that all aspects of binding, including specific saturable binding as well as nonspecific partitioning are captured. We extended the concentration range over five orders of magnitude and chose appropriate concentration ranges for each of the tested PFAS to capture all phases of the binding curves. The drawback of literature data was that it was typically measured at a narrower concentration range, which resulted in measurements at different phases of the binding curves, which limits comparisons of the experimental binding constants.

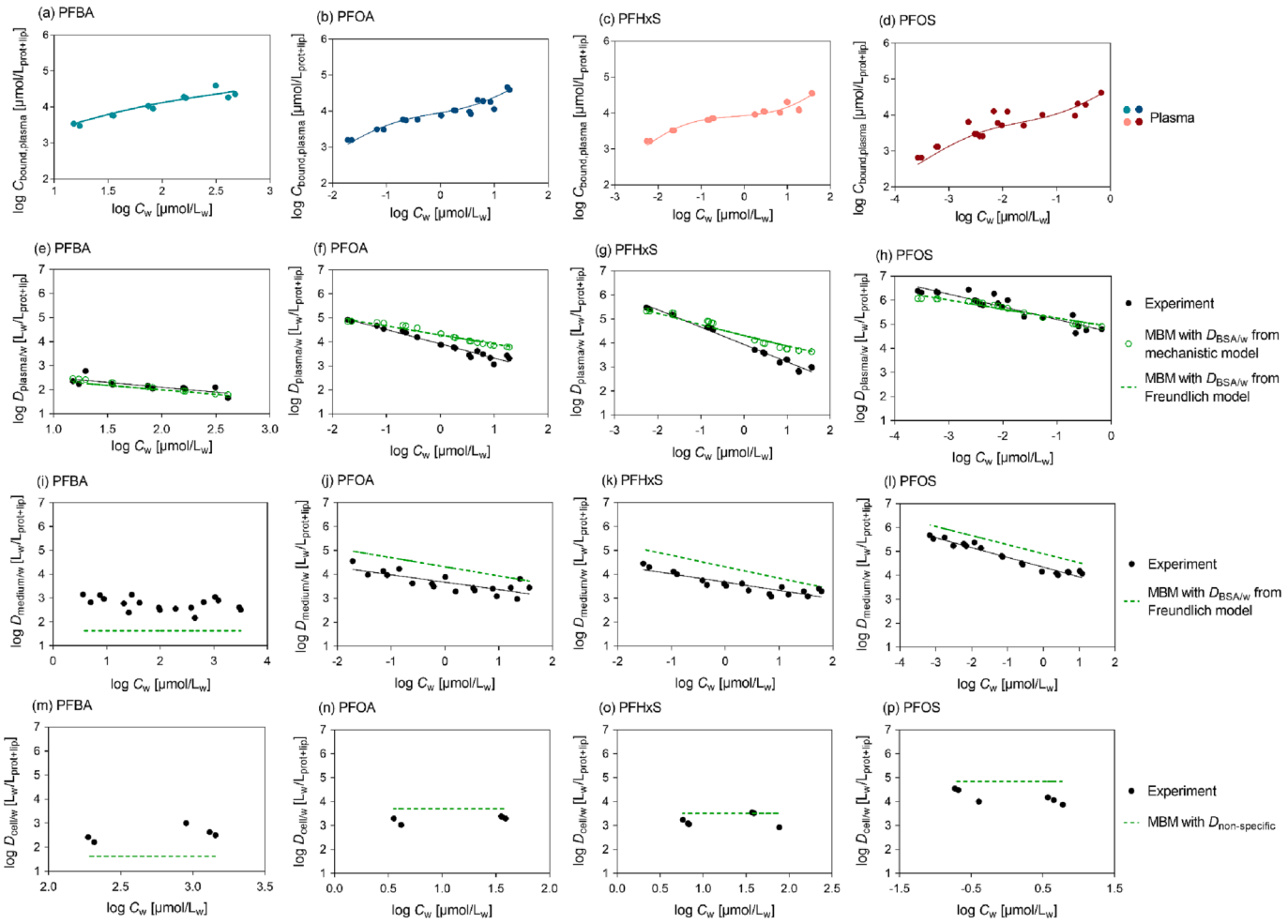


Fig. 3. Partitioning to biological materials (plasma, medium, cells). The concentration of (a) PFBA, (b) PFOA, (c) PFHxS and (d) PFOS bound to plasma ($C_{bound,plamsa}$, circles) were fitted with a mechanistic combined binding/partitioning model (Eq. (16)). Distribution ratios between plasma and water ($D_{plasma/w}$) of (e) PFBA, (f) PFOA, (g) PFHxS and (h) PFOS were fitted phenomenologically by the Freundlich-type model (Eq. (12), black solid line) and predicted by the mass balance model (MBM, Eq. (17)) with concentration-dependent BSA-water distribution ratios, $D_{BSA/w}$ from the Freundlich-type model (Eq. (12), Table 1, green dotted line) or from the mechanistic combined binding/partitioning model (Eq. (16), green hollow circle), and the liposome-water distribution ratio, $D_{lip/w}$ (Table 2). Distribution ratio between medium and water ($D_{medium/w}$) of (i) PFBA, (j) PFOA, (k) PFHxS and (l) PFOS were fitted phenomenologically by the Freundlich-type model (Eq. (12), black solid line) and predicted by the MBM (Eq. (17), green dotted line) with the $D_{BSA/w}$ from the Freundlich-type model (Eq. (12), Table 1) and the $D_{lip/w}$ (Table 2). Distribution ratios between cell and water ($D_{cell/w}$) of (m) PFBA, (n) PFOA, (o) PFHxS and (p) PFOS were calculated by Eq. (11) (black circle) from experiments and predicted by the MBM (Eq. (17), green dotted line) with $D_{non-specific}$ of BSA (Table 1) and $D_{lip/w}$ (Table 2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

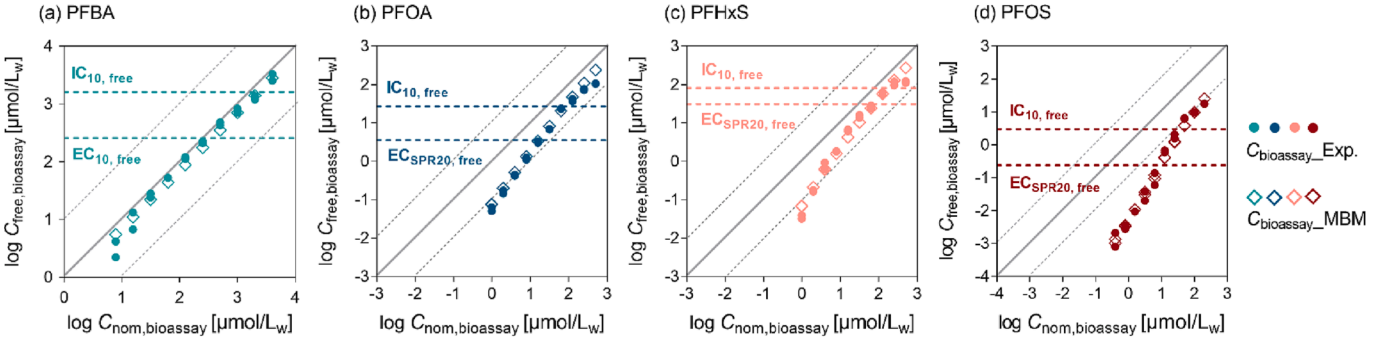


Fig. 4. Freely dissolved concentration of (a) PFBA, (b) PFOA, (c) PFHxS and (d) PFOS in GeneBLAzer medium (220 μL) in presence of 21,000 cells after 24 h cell exposure. log $C_{free,bioassay}$ of PFAS were quantified with SPME (Eq. (18), filled circles) and predicted by mass balance model (MBM, Eq. (20), empty diamond) with the concentration-dependent medium-water distribution ratio, $D_{medium/w}$ (Eq. (12), Table A6) and cell-water distribution ratio, $D_{cell/w}$ (Table 2). Inhibitory concentration ($IC_{10,}$ Eq. (22)) for cytotoxicity and effect concentration (EC_{10} in agonist mode (Eq. (23)) or EC_{SPR20} in antagonistic mode (Eq. (26))) were measured with the PPARγ-GeneBLAzer assays.

3.3. Binding to human plasma

C18-SPME was also used to measure C_{free} in plasma samples and to derive binding curves. As shown in Fig. 3a-e, the plasma binding curves of all four PFAS were also concentration-dependent and could be fitted by the mechanistic combined binding/partitioning model. To align the units, the saturation binding (Eq. (14)) was multiplied by the volume fraction of protein ($V_{prot}/V_{prot+lip} = 0.909$) in the plasma. $D_{plasma/w}$ ($\nu < 1$) in Table 2 refers to the specific binding constant $K_{specific}$ (Eq. (16) and was derived from the low-concentration portion of the binding curves with Eq. (14). $D_{plasma/w}$ ($\nu > 1$) refers to non-specific binding (Eq. (16)). All fit parameters of the mechanistic binding model are listed in Table A7.

The binding curves to BSA and to plasma overlapped for PFBA (Fig. A4a) and PFOS (Fig. A4d), but only overlapped at the low concentration range of protein-specific binding and gradually separated as the concentration increased for PFOA (Fig. A4b) and PFHxS (Fig. A4c). The good agreement is an indication that the plasma binding is dominated by protein binding with a low contribution of the lipids to the overall plasma binding, which justifies the application of the mechanistic combined binding/partitioning model that was derived for BSA.

Despite the non-linearity of plasma binding, we also fitted the experimental $D_{plasma/w}$ with the Freundlich-type model (Eq. (12)) and compared the $\log D_{plasma/w} - \log C_w$ relationship with the prediction by the MBM (Fig. 3e-h, Table 2, and Table A7). There was a decent agreement between the experimental $D_{plasma/w}$ (black circle) and the fit with the Freundlich-type model (black solid lines) and the MBM prediction that was based on the Freundlich model for $D_{BSA/w}$ (green dashed lines) as well as the MBM prediction that were based on the mechanistic protein binding model (green circles) in Fig. 3e-h.

The MBM for plasma based in the mechanistic combined binding/partitioning model for BSA was also used to analyze how lipids and proteins contribute to the bound PFAS. $D_{plasma/w}$ ($\nu < 1$) predicted for realistically low concentrations with $K_{specific_BSA}$ (Table 1) were comparable to the experimental ones with>98% contribution by protein binding (Table 2). This also rationalizes the overlay of the binding curves to BSA and plasma. $D_{plasma/w}$ ($\nu > 1$) predicted for high concentrations with $D_{non\text{-}specific_BSA}$ of PFBA, PFOA, PFOS agreed well with the experimental $D_{plasma/w}$ and the contribution of lipid increased from approximately 1% for low concentrations and specific binding (Table 2) to 10% for high concentrations and dominant non-specific binding (Table A7).

Forsthuber et al. (2020) analyzed 11 PFAS bound to albumin and lipoproteins in human blood samples and concluded that albumin was the main carrier in plasma. However, this study only analyzed the blood concentration and did not investigate any binding mechanism. C18-SPME combined with MBM can not only obtain partition constants, but also differentiate the specific and non-specific binding within wide concentration ranges and predict the contribution of lipids and proteins.

3.4. Partitioning in medium

We further studied the partitioning of PFAS in GeneBLAzer assay medium that is commonly used for reporter gene assays in Tox21. Binding curves are depicted in Figure A5. Volume fractions of protein and lipid in the GeneBLAzer assay medium were 0.08% and 0.0007% (Table A5). The $C_{bound,medium}$ of PFAS was also fitted with the combined binding/partitioning model (Eq. (16)). The saturation binding portion of Eq. (14) was multiplied by the volume fraction of protein ($V_{prot}/V_{prot+lip} = 0.992$) in the medium to align the units. Apart from PFOS (Figure A5d), which showed an excellent fit, the other three PFAS did not show the pronounced non-linear binding and a much lower quality of the fit (Figure A5 a-c). The highest concentration of PFOA and PFHxS were excluded from the fit, but the model was still not able to explain the shape of the binding curves. This may be caused by protein denaturation at high concentrations due to the surface-active properties of PFOA and PFHxS, which may have caused unfolding of the protein (MacManus-Spencer et al., 2010).

Since PFAS binding to protein was concentration-dependent and the protein content was 140 times higher than lipid content in the GeneBLAzer assay medium, the $\log D_{medium/w}$ of PFOA, PFHxS and PFOS (Eq. (12)) were also concentration-dependent. The $\log D_{medium/w}$ of PFOA, PFHxS and PFOS predicted with the MBM (Eq. (17), Table A6) showed the same trends as the experimental results but predictions resulted in slightly higher $D_{medium/w}$ than experiments (green dashed line in Fig. 3j-k). For the hydrophilic PFBA, the $\log D_{medium/w}$ was nearly constant over the tested concentration range and the predicted $D_{medium/w}$ of PFBA was lower than experimental ones (Fig. 3i), which might have been caused by the slightly higher recovery of the SPME (95–120%, Eq. (5)), resulting in lower $C_{bound,medium}$ and $D_{medium/w}$ calculated for the MBM (Eqs. (10) and (11)).

3.5. Partitioning to cells

The concentrations of PFAS bound to HEK293H cells ($C_{bound,cell}$, Eq. (10)) were measured at two nominal concentrations that differ by a factor of eight (Table A2). The volume fractions of water, protein and lipid of cells were 91.6%, 8.14% and 0.432% (Table A5), which were comparable to the values determined in previous study (Fischer et al., 2017). $D_{cell/w}$ derived from different concentrations of the same PFAS did not show any statistical difference (Student's t test, $p > 0.05$) and an average value of $\log D_{cell/w}$ for each PFAS is listed in Table 2.

A previous study pointed out that structural proteins (e.g. cytoskeleton) are abundant in mammalian cells and organic acids have lower affinity to structural proteins compared to BSA (Henneberger et al., 2016). Non-specific binding of PFAS to structural protein and lipid may dominate their partitioning to the cells. Based on this assumption, $D_{cell/w}$ was predicted by the MBM (Eq. (17)) with $D_{non\text{-}specific}$ (Table 1) as a surrogate for partitioning to structural proteins and $D_{lip/w}$ (Table 2) for lipid.

The predicted $D_{cell/w}$ (green dashed line in Fig. 3m-p) were not much different from the experimental $D_{cell/w}$ for PFOA, PFHxS and PFOS with discrepancies within 15%. For PFBA, the predicted $D_{cell/w}$ was around 35% lower than the experimental $D_{cell/w}$ similarly to what has been observed for medium. Here, we considered only proteins and lipids of the cell homogenate as the main sorptive phase for $D_{cell/w}$, which is different from Henneberger et al (2019a) where the cells were recognized as an entity containing water, protein and lipid and all of them contributed to cell-water distribution.

3.6. Freely dissolved concentration of PFAS in PPARγ reporter gene assays

In the bioassay, the partitioning of PFAS to medium components and cells inevitably resulted in substantial differences between freely dissolved concentration ($C_{free,bioassay}$) and nominal concentration ($C_{nom,bioassay}$). The $\log D_{f/w}$ in Table A6 were used to derive $C_{free,bioassay}$ in complex assay medium from measured concentrations in the fiber by Eq. (18). As shown in Fig. 4, if $C_{free,bioassay}$ is equal to $C_{nom,bioassay}$, data points fell on the 1:1 diagonal. For the hydrophilic PFBA, $\log C_{free,bioassay}$ and $\log C_{nom,bioassay}$ were almost equal with deviations below 10%, but for hydrophobic PFHxS, PFOA and PFOS, $C_{free,bioassay}$ were up to 100 times lower than $C_{nom,bioassay}$. Our results are consistent with a recent study that hydrophilic chemicals and highly protein-bound chemicals behave differently on C_{free} (Dimitrijevic et al., 2022).

Considering medium components and cells may be contributing to PFAS depletion in bioassays, a MBM (Eq. (20)) with $D_{medium/w}$ and $D_{cell/w}$ for these PFAS (Table A6 and Table 2) was used to predict the values of $C_{free,bioassay}$ of PFAS. The $\log C_{free,bioassay}$ predicted with the MBM agreed well with experimental data (Fig. 4). This means that existing PFAS effect data can be assessed for their freely dissolved effect concentrations in the future by the simple MBM, provided that the distribution ratios ($D_{BSA/w}$ and $D_{lip/w}$) and experimental conditions (lipid and protein content of medium and cells) are available.

Table 3
Nominal and freely dissolved concentration of PFBA, PFOA, PFHxS and PFOS in PPARγ reporter gene assays and human plasma. Quantitative *in vitro* to *in vivo* extrapolation (QIVIVE) ratio for workers and residents living near fluorochemical plants and general population.

	PPARγ reporter gene assay			Workers or residents near fluorochemical plant		General residents	
	Cytotoxicity	PPARγ-activation		Human plasma		Human plasma	
	$IC_{10,nom}$ [μmol/L_w]	Agonism $EC_{10,nom}$ [μmol/L_w]	Antagonism $EC_{SPR20,nom}$ [μmol/L_w]	$C_{nom,plasma}$ [μmol/L]	$QIVIVE_{nom}$ ratio	$C_{nom,plasma}$ [μmol/L]	$QIVIVE_{nom}$ ratio
PFBA	2.20×10^3	316		0.408[a]	1.29×10^{-3}	5.30×10^{-3}[c]	1.68×10^{-5}
PFOA	91.0		18.2	35.8[b]	1.96	1.23×10^{-2}[d]	6.74×10^{-4}
PFHxS	125		70.7	26.4[b]	0.373	1.40×10^{-2}[d]	1.98×10^{-4}
PFOS	29.5		5.54	126[b]	22.7	3.30×10^{-2}[d]	5.95×10^{-3}
	$IC_{10,free}$ [μmol/L_w]	Agonism $EC_{10,free}$ [μmol/L_w]	Antagonism $EC_{SPR20,free}$ [μmol/L_w]	$C_{free,plasma}$ [μmol/L]	$QIVIVE_{free}$ ratio	$C_{free,\ plasma}$ [μmol/L]	$QIVIVE_{free}$ ratio
PFBA	1.61×10^3	259		5.72×10^{-2}	2.21×10^{-4}	7.42×10^{-4}	2.87×10^{-6}
PFOA	27.0		3.64	1.93×10^{-2}	5.31×10^{-3}	6.63×10^{-6}	1.82×10^{-6}
PFHxS	81.6		31.9	3.76×10^{-3}	1.18×10^{-4}	2.00×10^{-6}	6.25×10^{-8}
PFOS	2.98		0.243	2.94×10^{-3}	1.21×10^{-2}	7.71×10^{-7}	3.17×10^{-6}

[c] Data from Gao et al. 2019.
[d] Data from https://www.cdc.gov/exposurereport/data_tables.html.
[a] Data from Bao et al. 2022.
[b] Data from Gao et al. 2015.

3.7. Effect concentrations of PFAS in PPARγ reporter gene assays

All PFAS were cytotoxic at the highest exposure concentrations (Figure A6). Inhibitory concentration that caused 10% of cytotoxicity ($IC_{10,free}$) (Eqs. (21) and (22)) of PFBA, PFOA, PFHxS and PFOS were 1610, 27.0, 81.6 and 2.98 μM respectively (Table 3). To avoid artefacts on the PPARγ activation caused by the cytotoxicity, the cellular effects were evaluated in the non-cytotoxic concentration range below IC_{10}.

Both agonistic and antagonistic effects in the PPARγ-GeneBLAzer reporter gene assay were observed after 24 h exposure to PFAS (Figure A6). PFBA showed agonistic effects up to 30% at lower concentrations and then suppression at higher concentrations, which were already impacted by cytotoxicity (Figure A6a). The effective concentration that triggered 10% of the maximum PPARγ activation ($EC_{10,free}$) (Eqs. (23) and (24)) of PFBA was 259 μM.

PFOA, PFHxS and PFOS showed antagonistic effects on PPARγ (Figure A6b-d), that were measured at a background of 80% activation effect of rosiglitazone. PFHxS suppressed nearly half of the signal from rosiglitazone, and PFOA and PFOS almost eliminated the signal. Suppression ratios (SPR) (Eq. (25)) of 20% of PFHxS, PFOA and PFOS were observed at $EC_{SPR20,free}$ of 3.64, 31.9 and 0.243 μM (Eq. (26), Table 3).

PPARγ-regulated signal pathways have been demonstrated to involve in PFAS disrupting lipid metabolism in many studies (Abbott 2009; Ma et al., 2018; Pan et al., 2021). Some researchers further investigated the interaction of PFAS with PPARγ by using reporter gene assays but their activities on the receptor were inconsistent. For example, no significant activation of PPARγ by PFOA and PFOS were observed on Cos-1 cells (Takacs and Abbott, 2007), which were transfected with plasmid DNA using lipofectamine reagent, while other studies proved they were potential agonists of PPARγ using transient transfected 3 T3-L1 cells (Vanden Heuvel et al., 2006) or HEK293 cells (Li et al., 2019, 2018; Zhang et al., 2014). PFOA, PFHxS and PFOS showed very faint activation effects in the PPARγ GeneBLAzer assay at the low concentration range in absence of rosiglitazone (data not shown) but obvious suppression without prior increase of the rosiglitazone signal when co-exposed with rosiglitazone.

3.8. Freely dissolved concentration of PFAS in human plasma

Many PFAS have been detected at micromolar levels in human plasma of firefighters and workers engaged in PFAS utilization and manufacturing facilities (Costa et al., 2009; Gao et al., 2015; Rotander et al., 2015; Sakr et al., 2007; Wang et al., 2012). For example, the maximum level of PFOA, PFHxS and PFOS of workers from one of the largest PFAS-related producers in China (Henxin Chemical Plant) were 35.8, 26.4 and 126 μmol/L (Table 3) (Gao et al., 2015). The hydrophilic PFBA was seldomly measured in plasma samples, but it was one of the dominant PFAS in urine (Peng et al., 2021). However, some studies applied more sensitive methods and found PFBA in maternal and cord serum (Bao et al., 2022; Gao et al., 2019). The maximum level of PFBA was 0.408 μmol/L among 50 pregnant women living near the fluorochemical industrial plant-Fuxin in China (Bao et al., 2022).

PFAS were also found in the general population at nanomolar level according to national health surveys among developing and developed countries (Cakmak et al., 2022; Gockener et al., 2020; Lin et al., 2019; Petriello et al., 2022; Yu et al., 2020; Zhang et al., 2019). The U.S. Center for Disease Control and Prevention conducted PFAS monitoring of the general U.S. population and their latest results (2017–2018) showed that the maximum concentration of PFHxS, PFOA and PFOS were 12.3, 14.0, 33.0 nmol/L respectively (https://www.cdc.gov/exposurereport/data_tables.html). The maximum concentration of PFBA was 5.30 nmol/L among 132 non-occupationally exposed pregnant women in Beijing, China (Gao et al., 2019).

$C_{free,plasma}$ was calculated with Eq. (27). Since the ν ($mol_{PFAS}/mol_{protein}$) of PFAS in human plasma samples were usually ≪1 and most of PFAS may bind highly specifically to proteins, leading to very low free PFAS concentrations.

$$C_{free,plasma} = \frac{C_{nom,plasma} \times V_w}{V_w + D_{plasma/w} \times V_{prot+lip,plasma}} \quad (27)$$

We derived the log $D_{plasma/w}$ at $\nu < 1$ (Table 2) from the binding curves of PFAS to human plasma (Fig. 3a-d) and used these values to calculate $C_{free,plasma}$ by the MBM (Eq. (27)). As shown in Table 3, $C_{free,plasma}$ was predicted to be lower than $C_{nom,plasma}$ but the ratio was highly chemical-specific. $C_{free,plasma}$ of PFBA was only seven times lower than $C_{nom,plasma}$ but this ratio increased with protein binding affinity to almost 40,000 for PFOS. There was little difference in this ratio for each PFAS between the exposed workers and the general population.

3.9. Quantitative in vitro to in vivo extrapolation

The QIVIVE ratio is the ratio of concentration of PFAS in human plasma and effect concentration EC (EC_{10} or EC_{SPR20}) in the bioassay (Henneberger et al., 2021). The $QIVIVE_{nom}$ ratio was calculated with Eq.

(28). Nominal concentration of PFAS in human plasma samples ($C_{nom, plasma}$) were collected from epidemiological studies and EC_{nom} were measured as illustrative example with the PPARγ GeneBLAzer assay.

$$QIVIVE_{nom}\ ratio = \frac{C_{nom,\ plasma}}{EC_{nom}} \quad (28)$$

The corresponding $QIVIVE_{free}$ ratio was calculated with Eq. (29) from $C_{free,plasma}$ (Eq. (27) and EC_{free} (calculated with Eq. (18)).

$$QIVIVE_{free}\ ratio = \frac{C_{free,plasma}}{EC_{free}} \quad (29)$$

As an illustrative example, we calculated the QIVIVE ratio for plasma concentrations (C_{plasma}) of PFBA, PFOA, PFHxS and PFOS in workers and residents living near fluorochemical plants from Bao et al. (2022) and Gao et al. (2015) or the general population in the US (https://www.cdc.gov/exposurereport/data_tables.html) and Gao et al. (2019), and the EC of PFAS in the PPARγ GeneBLAzer assay (Table 3).

A QIVIVE ratio > 1 would indicate that the particular *in vitro* effect on the PPARγ receptor may have the potential of causing associated adverse outcomes in *in vivo* (Sipes et al., 2017). Notably, this assay is only one example - for a full QIVIVE, one needs to measure many *in vitro* endpoints and evaluate the distributions of QIVIVE ratios. This can be done in the future when more bioassay data become available from HTS studies.

The $QIVIVE_{nom}$ ratios of PFOA and PFOS were 1.96 and 22.7, suggesting PFOA and PFOS might cause an *in vivo* effect after occupational exposure with respect to this particular biological endpoint (Table 3, Fig. 5). In epidemiological studies, associations between decrease of PFOS and PFOA concentrations in serum and low-density lipoprotein cholesterol have been demonstrated (Fitz-Simon et al., 2013), which hints at the relevance of lipid metabolism and hence binding to PPARγ as relevant molecular initiating event.

The $QIVIVE_{free}$ ratio of PFOA and PFOS were reduced to 5.87×10^{-3} and 1.34×10^{-2} (Table 3, Fig. 5), indicating that proteins and lipids in plasma may act as reservoirs of PFAS, reducing the freely dissolved concentration of PFAS. But this does not mean it is safe since a majority of bound PFAS still accumulate in the body and can be remobilized. The large difference between $QIVIVE_{nom}$ and $QIVIVE_{free}$ for anionic PFAS is because protein binding is highly concentration-dependent with a high-affinity specific binding at low concentrations and lower non-specific binding at higher concentrations. Thus, the low concentrations in human plasma are strongly bound to proteins, while the high concentration needed to trigger effects in *in vitro* bioassays have a higher fraction unbound.

Since the concentrations of these PFAS in general population were thousands of times lower than those in workers or residents living near fluorochemical plants, both $QIVIVE_{nom}$ and $QIVIVE_{free}$ ratios of general population were thus calculated to be thousands lower (Table 3, Fig. 5).

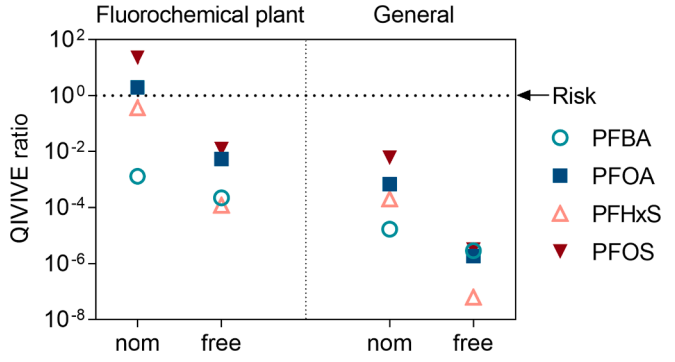


Fig. 5. Quantitative *in vitro* to *in vivo* extrapolation (QIVIVE ratio) for workers and residents living near fluorochemical plants and general population based on nominal concentrations (nom, (Eq. (28)) or free concentrations (free, (Eq. (29)) of human exposure level and effective concentrations in the PPARγ-GeneBLAzer bioassay.

4. Conclusion

The use of PFAS has improved the quality of our life, but their persistence and continued usage perpetuate their adverse impact on humans and the environment. HTS cell-based bioassays can be used to characterize the potential toxicity of PFAS, and the associated QIVIVE ratios allow one rank their relative risk (Bell et al., 2018). This may be a practical strategy to deal with a conflict between commercial needs and environmental health protection.

PFAS bound to proteins in plasma can also be transported to cells or different tissues in bound form with the blood and released at other organs and sites. There is an equilibrium distribution of C_{free} and C_{bound} (Fischer et al., 2018) due to the reversible binding of PFAS to proteins. Receptors with high affinity could trigger the dissociation from serum albumins acting as transporters. Such redistribution can occur in blood and in the *in vitro* bioassays and therefore C_{free} can be considered still a useful parameter from comparisons.

The binding of PFAS to proteins is strongly concentration-dependent. All biomaterials that have much higher protein than lipid content showed also concentration-dependent binding curves, which is a great experimental challenge if many anionic PFAS need to be characterized. Given the small contribution of lipid binding in case of human plasma, it is possible to use $D_{BSA/w}$ as a surrogate for $D_{plasma/w}$ but for plasma of other environmental organisms with higher lipid content, lipid partitioning would need to be included.

The MBM yielded an adequate prediction of C_{free} in cell-based bioassays. Although we demonstrated the validity of the MBM only for four anionic PFAS, one type of bioassay medium and one cell line, previous work on other organic anions has demonstrated that models can be transferred to other bioassay setups if the lipid and protein content has been quantified (Henneberger et al., 2019b; Huchthausen et al., 2020). Again, it must be stressed that the nonlinear binding behavior of anionic PFAS to proteins is not a feature of the perfluorination of the molecules but caused by the deprotonated carboxylic acid or sulfonic acid group of the tested PFAS. Thus, with respect to protein binding, our results and comparison to protein binding of non-fluorinated organic ions demonstrate that the anionic PFAS behave like other organic anions.

The PPARγ reporter gene assay is just one exemplary HTS assay that was included in Tox21. The Tox21 collaboration has employed>60 HTS assays for chemical assessments, including assays for cytotoxicity, cell stress, mitochondrial function and nuclear receptor activation (Sipes et al., 2017). A bioassay test battery that covers a range of relevant modes of action may be used for overall toxicity evaluation for commercially available PFAS. 75 PFAS were selected for high-throughput effect screening (Patlewicz et al., 2019) conducted by researchers of U.S. EPA and the National Toxicology Program (https://www.epa.gov/pfas, accessed January 2023) and the results are still pending. Furthermore, more complex *in vitro* models such as models derived from human induced pluripotent stem cells (Corbett 2019) may serve in the future as a more suitable point of departure for QIVIVE.

As the anionic PFAS have shown such a strong and non-linear binding to proteins, their true internal effect concentrations might be much different from neutral PFAS and the toxic effects of hydrophobic anionic PFAS may be underestimated as compared to the hydrophilic ones. Therefore, $C_{free,bioassay}$ may be a more appropriate dose-metric for comparative effect assessment and QIVIVE of PFAS. The MBM was successfully used to predict $C_{free,bioassay}$ of PFAS in one bioassay, which will greatly improve the accuracy of HTS to identify specific effects and compare effects across bioassay types. It will not always be necessary to measure C_{free} of PFAS but existing HTS effect data based on nominal concentrations can be converted to free effect concentrations using the MBM.

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CRediT authorship contribution statement

Weiping Qin: Conceptualization, Investigation, Validation, Formal analysis, Writing – original draft, Writing – review & editing, Funding acquisition. **Luise Henneberger:** Methodology, Validation, Writing – review & editing. **Julia Huchthausen:** Methodology, Validation, Writing – review & editing. **Maria König:** Investigation, Writing – review & editing. **Beate I. Escher:** Conceptualization, Methodology, Validation, Visualization, Resources, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary material

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